Custom & Suggested Itinerary Workflows: End-to-End Analysis & Proposal

All Use Cases Evaluated (User & Developer Perspectives) • Live Target: edselrenz.github.io/Itinerary-App/tabs/home • Date: November 6, 2026

1. Executive Summary: Eleven Use Cases Audited Across Both Planning Engines

We conducted an exhaustive static code audit of the uploaded monorepo (`Project/` built on Angular 22 and Ionic 9; `backend/` built on FastAPI and SQLAlchemy 2.0) paired with live interactive testing on https://edselrenz.github.io/Itinerary-App/tabs/home covering every use case of both the **Suggested Itinerary Wizard** (/suggested-wizard) and the **Custom Itinerary Builder** (/custom-builder, `/custom-builder/:id`, and /itinerary/:id).

While both workflows share a clean 3-stage conceptual progression (*Questionnaire → Place Selection / Clustering → Day Timeline & Persistence*) and implement honest null-cost budget rollups via `ItineraryService.calculateBudgetSummary()`, live testing and code tracing exposed **12 concrete functional and UI/UX defects**—including two critical data-corruption bugs in Guest/Offline mode:

- **Critical Metadata Loss When Saving Custom Itineraries in Guest/Offline Mode (`C-BUG-06`):** In `custom-builder.page.ts:806–826` and `1114–1126`, `addPlaceStop()` and `saveItinerary()` omit `custom_title: it.name` and `place` from the save payload. When saved via `GuestStorageService` (`createGuestItinerary()` at `itinerary.service.ts:421–435`), every catalog stop is persisted with `place: null` and `custom_title: undefined`—causing every stop in a saved Custom Itinerary to render on `/itinerary/:id` as **"Destination Stop — Explore the river and limestone formations..."** with a fabricated `₱200` fee and zero coordinates.
- **Dead `matchedPool` Variable in Offline Suggested Generator (`S-BUG-03`, Critical):** In `itinerary.service.ts:753–796`, `generateOfflineSuggestedItinerary()` filters and sorts `matchedPool` by the user's chosen municipality and preferences on lines `753–771`, **but never reads `matchedPool` afterwards**. Selecting *Pantabangan*, *Cabanatuan City*, *Gapan City*, or *General Tinio* offline sets `isProvincialOrBongabon = false` and falls back to `[curatedPool[0], curatedPool[1], curatedPool[2]]`, scheduling the exact same 3 Bongabon venues on every single day of the trip.
- **Unconditional 4-Second Offline Network Stall on Suggested Generation (`S-BUG-04`, High):** Even when `ConnectivityService.isHostOffline$` is already `true`, `ItineraryService.suggestItinerary()` fires a live HTTP `POST` and waits for a 4,000 ms timeout before invoking the local generator, leaving the *"Clustering recommendations..."* spinner hanging in offline sessions.
- **Divergent Dual Editors & Missing Form Controls (`C-BUG-01` to `C-BUG-05`, High):** Step 1 of `CustomBuilderPage` omits the `Trip Title` input, quick destination chips, and budget presets; Step 2's category filter drops 7 of the 16 canonical POI categories; changing dates in Step 3's settings card fails to sync the `days` array; and `ItineraryDetailPage` introduces a competing in-place editor that orphans `/custom-builder/:id` and collapses min/max cost ranges into a single number.

2. Complete Use-Case Analysis: Suggested vs. Custom Itinerary

The table below documents all 11 use cases across both workflows, detailing the execution path in code, verified live behavior on edselrenz.github.io/Itinerary-App/tabs/home, and specific defects uncovered.

Use Case	Workflow Stage & Code Path	How It Works (Live & Code Trace)	Identified Bugs & Edge Cases
UC-S1 Suggested Entry & Deep-Link	`HomePage` / `MapPage` → `/suggested-wizard?destination=...&placeId=...`	Launched from Home's *Suggested Itinerary* card, Home's *Build Itinerary Here* modal CTA, or Map's *Start Trip Here* bottom-sheet CTA.	**S-BUG-01:** `suggested-wizard.page.ts:171` reads `params['destination']` but ignores `params['placeId']`, never anchoring or guaranteeing the clicked POI in the generated plan.
UC-S2 Suggested Form & Date Modal	`SuggestedWizardPage` (`viewMode === 'form'`, L32–132)	Collects `destination` (6 quick chips), `start_date`/`end_date` via pop-up modal (`2D1N`, `3D2N`, `4D3N`, `1 Week`), `daily_budget` (4 presets), and 5 preference pills.	**S-BUG-02:** `openDatePicker()` (L190) and `setPresetDuration()` (L207) still use UTC `new Date(str).toISOString()` instead of `parseLocalIsoDate()`, risking off-by-one shifts in UTC+8.
UC-S3 Suggested Generation Engine	`ItineraryService.suggestItinerary()` → `itinerary_generator.py` or `generateOfflineSuggestedItinerary()`	Online: SQL bbox pre-filter + Haversine scoring (`25 km` radius, `+40` pref, `+20` featured, `-25` dup cat, cumulative budget) + chronological sort. Offline: local TypeScript generator.	**S-BUG-03 (Critical):** Offline `matchedPool` (L753–771) is never used; non-Bongabon towns repeat the same 3 Bongabon stops every day. **S-BUG-04:** Waits 4s for HTTP timeout even when offline banner is active.
UC-S4 Suggested Preview & Stop Modal	`SuggestedWizardPage` (`viewMode === 'itinerary'`, L134–285)	Renders Trip Card, `Verified Estimated Costs`, collapsible `Trip Assumptions` insight cards, segmented Day capsule bar, photo stop cards, and Stop Detail modal.	No route map or polyline preview; Day 2 & Day 3 offline presets zig-zag `176.2 km` across the province (General Tinio → Pantabangan → Gapan City).
UC-S5 Suggested Nearby Essentials	`SuggestedWizardPage` (`viewMode === 'nearby'`, L287–360)	Toggles 3-tab view (*Restaurants*, *Hospital & Pharmacies*, *Stores*) filtered by active day's municipality; `+ Add to Day X` appends stop.	**S-BUG-05:** `addNearbyPlaceToActiveDay()` (L422) hardcodes `12:00 PM`, Bongabon coords (`15.63, 121.14`), and `null` cost, and appends after overnight stays without re-sorting.
UC-S6 Save vs. Customize Handoff	`acceptAndSave()` & `customizeInBuilder()` (L475–555)	*Accept & Save* persists trip and routes to `/tabs/home`. *Customize in Builder* saves trip and routes to `/custom-builder/:id`.	**S-BUG-06:** `customizeInBuilder()` writes a permanent copy to storage *before* opening the builder; cancelling in the builder leaves an unwanted duplicate saved trip.
UC-C1 Custom Step 1 Setup Form	`CustomBuilderPage` (`activeView === 'form'`, L48–137)	Collects Destination, Travel Dates (via pop-up modal), Daily Budget, and Preferences, then calls `proceedToBuilder()` to sync `days` array.	**C-BUG-01:** Missing `Trip Title` input field in Step 1 HTML (defaults silently to *"Nueva Ecija Getaway"*); missing quick destination & budget chips; button mislabeled *"Find Itinerary"*.

Use Case	Workflow Stage & Code Path	How It Works (Live & Code Trace)	Identified Bugs & Edge Cases
UC-C2 Custom Step 2 Browse & Add	`CustomBuilderPage` (`activeView === 'browse'`, L138–266)	Target day selector chips (`Day 1 (N)`, +), 300ms debounced server/offline search, 5 category pills, 2-col place grid with `Add` / `Added` toggle.	**`C-BUG-02`:** `catMap` (L697) maps only 9 of 16 categories, dropping 7 categories. **`C-BUG-03`:** Ignores Step 1 `destination` and overwrites multi-preferences with a single tab. **`C-BUG-04`:** Cards lack detail modal inspection.
UC-C3 Custom Step 3 Timeline & Edit	`CustomBuilderPage` (`activeView === 'timeline'`, L268–475)	Trip settings card, budget summary, day tabs, numbered stop timeline with ↑/↓ reorder, `Edit Scheduled Stop` modal (time, dur, cost, notes, move-to-day), and delete confirmation.	**`C-BUG-05`:** Editing `start_date` / `end_date` in the timeline settings card does not resize `this.days` or update `day.travel_date`. **`C-BUG-06` (Critical):** Omitting `custom_title: it.name` and `place` turns saved guest stops into *"Destination Stop"*.
UC-C4 Place Picker & Off-Grid Venue	`CustomBuilderPage` (`isPlacePickerOpen`, L562–835)	2-tab modal: *Browse Places* (live search) and *Off-Grid Venue* (custom title, address, min/max cost, planned time, optional NE lat/lng bounds check).	Clicking *"Add Off-Grid Custom Venue"* opens `pickerTab = 'browse'` instead of `'custom'` (L735); `[disabled]="isPlaceInActiveDay(id)"` makes duplicate confirmation alert dead code.
UC-C5 Saved Trip Editing & Guards	`pendingChangesGuard`; `/custom-builder/:id` vs. `/itinerary/:id`	`CustomBuilderPage` enforces `canDeactivate` guard. `ItineraryDetailPage` provides an inline `isEditMode` form.	Competing dual editors: `/itinerary/:id` inline editor orphans `/custom-builder/:id`, lacks `canDeactivate` guard, has no catalog picker, and collapses `min/max` cost ranges into one number.

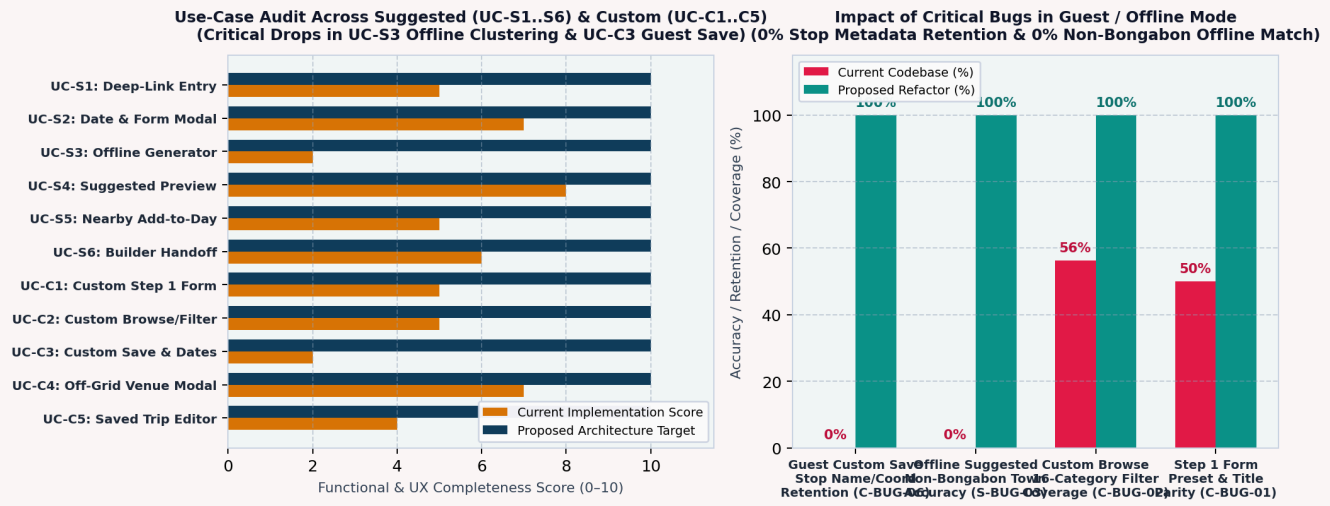


Figure 1: Left: Functional & UX health evaluation across all 11 use cases (`UC-S1`–`UC-S6` and `UC-C1`–`UC-C5`) comparing current implementation against the proposed architecture. Right: Quantitative impact of current bugs in Guest/Offline mode—specifically the 0% catalog stop metadata retention on saved Custom Itineraries (`C-BUG-06`), the 0% non-Bongabon stop accuracy in offline Suggested trips (`S-BUG-03`), and the 7 unmapped OSM categories in Custom Browse (`C-BUG-02`).

3. Root-Cause Code Evidence for the Two Critical Data-Corruption Bugs

Before presenting the UI/UX recommendations, it is essential for the engineering team to see the exact lines of TypeScript causing the two highest-severity defects in Guest/Offline mode.

3.1 Critical Bug C-BUG-06: Why Saved Custom Itineraries Lose All Place Names, Photos & Coordinates

When a user adds a catalog destination in `CustomBuilderPage.addPlaceStop()` (`custom-builder.page.ts:806–826`), the stop stores `place_id: place.id` and `name: place.name`, leaving `custom_title` undefined. When `saveItinerary()` (lines `1114–1126`) serializes the payload, it drops `it.name` and `place`:

```
// 1. custom-builder.page.ts (Lines 1114–1126) — Drops it.name and place object!
items: d.items.map((it) => ({
  place_id: it.place_id,
  custom_title: it.custom_title, // undefined for all catalog places!
  custom_lat: it.custom_lat,     // undefined for all catalog places!
  custom_lng: it.custom_lng,     // undefined for all catalog places!
  sort_order: it.sort_order,
  planned_time: it.planned_time,
  duration_minutes: it.duration_minutes,
  notes: it.notes,
  estimated_cost_min: it.estimated_cost_min,
  estimated_cost_max: it.estimated_cost_max,
}))

// 2. itinerary.service.ts (Lines 421–435) — Saves place: null in GuestStorage!
const items: ItineraryItem[] = (d.items || []).map((it, itIdx) => ({
  id: this.generateId(),
  day_id: dayId,
  place_id: it.place_id,
  custom_title: it.custom_title, // undefined!
  ...
  place: it.place || null,       // null!
}));

// 3. itinerary-detail.page.ts (Lines 670–672) — Renders "Destination Stop" fallback!
public getStopTitle(item: ItineraryItem): string {
  return item.place?.name || item.custom_title || 'Destination Stop';
}
```

3.2 Critical Bug S-BUG-03: Why Offline Suggested Itineraries Ignore Chosen Municipalities & Preferences

In `ItineraryService.generateOfflineSuggestedItinerary()` (`itinerary.service.ts:753–796`), the developer wrote logic to filter and sort `matchedPool` for `CR-009`, but never connected `matchedPool` to `dayPlaces`:

```
// Project/src/app/core/services/itinerary.service.ts (Lines 753–796)
let matchedPool = curatedPool;
if (targetDest && !targetDest.includes('nueva ecija')) {
  const muniMatches = curatedPool.filter((p) => p.municipality.toLowerCase().includes(targetDest));
  if (muniMatches.length > 0) {
    matchedPool = muniMatches; // Filtered & sorted on lines 753–771, but NEVER READ AGAIN!
  }
}
...
const isProvincialOrBongabon = !targetDest || targetDest.includes('nueva') || targetDest.includes('ecija')
|| targetDest.includes('bongabon');
const cluster = isProvincialOrBongabon ? dayClusterPresets[d % dayClusterPresets.length] : null;

// BUG: When user selects "Pantabangan" or "Gapan City", cluster is null, so it schedules
// curatedPool[0..2] (Rohi Cafe, Museo de Bongabon, Thornes BBQ in Bongabon) on EVERY day!
const dayPlaces = cluster?.places || [
  curatedPool[0],
  curatedPool[1],
  curatedPool[2],
];
```

4. User's Point of View: Traveler UI/UX Critique & Suggestions

Evaluated against the W3C WCAG 2.2 AA usability standards and domain best practices for mobile trip planners, travelers encounter six major UX friction points across the Custom and Suggested workflows:

1. **Identical Step 1 Questionnaires Blurred Together (*"Am I in Suggested or Custom?"*):** Both /suggested-wizard and /custom-builder open with the exact same headline (*"Tell us your travel details"*) and the exact same bottom CTA button label (**"Find Itinerary"**). Yet on Custom Builder Step 1, the user cannot even enter a **Trip Title**, nor do they get the quick municipality or budget chips available in Suggested Wizard!
   User UX Suggestion: Differentiate the two entry screens clearly. On **Custom Builder Step 1**, lead with a prominent **Trip Title** input (e.g., *"Weekend Family Roadtrip"*), include the Quick Destination & Budget preset chips, and rename the primary button to **"Next: Choose Places to Visit →"**.
2. **"Add Off-Grid Custom Venue" Button Opens the Wrong Modal Tab:** In Step 3 of the Custom Builder, tapping the button explicitly labeled **"Add Off-Grid Custom Venue"** opens the Place Picker modal on the **Browse Places** tab instead of the **Off-Grid Venue** form, forcing the user to hunt for the secondary segment tab at the top of the sheet.
   User UX Suggestion: Pass the target tab directly into `openPlacePicker('custom')` so tapping *Add Off-Grid Custom Venue* opens the custom venue form immediately.
3. **Cannot Inspect a Destination Before Adding It in Custom Browse Mode:** In Step 2 of /custom-builder (`activeView === 'browse'`), place cards only show a thumbnail, name, municipality, cost tag, and `+ Add` button. Tapping a card's photo or title does nothing—travelers cannot read the description, check opening hours, or see commute info without leaving the Custom Builder!
   User UX Suggestion: Make the card body/photo clickable in Browse mode to open the shared **Stop Detail Modal**, with a primary **"+ Add to Day {{ activeDayNumber }}"** CTA inside the modal.
4. **No Auto-Calculated Schedule Times or Transit Connectors in Custom Builder:** When a user adds 3 places to Day 1 in Custom Builder, every stop card displays only `(2 hrs)` in amber with no start/end time until the user manually edits each stop one by one, and there is no warning if a user schedules a stop outside its verified opening hours (e.g., scheduling a night market at 9:00 AM).
   User UX Suggestion: Automatically chain default sequential time windows (e.g., Stop 1: `09:00 AM - 10:30 AM`, Stop 2: `11:00 AM - 12:30 PM` with a 30-min transit buffer) as stops are added or reordered, and add a one-tap **"Auto-Optimize Day Route & Times"** button on the timeline header.
5. **Confusing Dual Save Buttons & Premature Saving on "Customize in Builder":** In Custom Builder's Timeline view, the top-right header says `✓ Save` while the bottom bar says `View Full Itinerary / Save Trip`, and both immediately exit to /tabs/home rather than opening the saved itinerary! Meanwhile, clicking **Customize in Builder** on a Suggested Itinerary saves a permanent copy to storage before the user even makes an edit.
   User UX Suggestion: Pass unsaved suggested drafts to Custom Builder via in-memory router state (saving only when the user clicks **Save Itinerary**), and redirect post-save to `/itinerary/:id` so the user sees their finished trip immediately.
6. **Two Different Editors for Saved Trips with Mismatched Capabilities:** Tapping `Edit` on /itinerary/:id opens a raw text-input form where adding an activity creates a blank text box (with no way to pick a real place from the database!), whereas `/custom-builder/:id` has the full visual timeline, catalog picker, and off-grid coordinate validator but is hidden from the UI.
   User UX Suggestion: Consolidate onto a single, unified visual editor (or add a **"Browse & Pick from Catalog"** button inside `/itinerary/:id` edit mode) so users never have to type destination names and descriptions by hand.

5. Developer's Point of View: Architecture & Code Improvement Proposal

To resolve all 12 functional and UI/UX defects cleanly while adhering to `coding-and-system-development` principles (SOLID, DRY, strict input validation, and minimal blast radius), we propose the following six engineering refactors:

Proposal 1: Fix Guest/Offline Custom Stop Serialization & Hydration (Fixes `C-BUG-06`)

Preserve the full `Place` reference on `BuilderStop` in `custom-builder.page.ts`, populate `custom_title: it.name` as a safe display fallback, and hydrate any stop with a `place_id` against `PlaceService` in `ItineraryService.mapGuestItinerary()`:

```
// Project/src/app/features/itinerary/custom-builder/custom-builder.page.ts
// 1. In BuilderStop interface & addPlaceStop(place: Place):
const stop: BuilderStop = {
  id: this.generateTempId(),
```

```
  place_id: place.id,
  custom_title: place.name,
  custom_lat: place.latitude,
  custom_lng: place.longitude,
  name: place.name,
  place: place, // Preserve full Place entity for offline/guest persistence
  ...
};

// 2. In saveItinerary() payload mapping:
items: d.items.map((it) => ({
  place_id: it.place_id,
  custom_title: it.custom_title || it.name,
  custom_lat: it.custom_lat ?? it.place?.latitude ?? null,
  custom_lng: it.custom_lng ?? it.place?.longitude ?? null,
  ...
  place: it.place || null,
}))
```

Proposal 2: Wire `matchedPool`, Budget Filtering & `placeId` Anchor in Offline Suggested Generator (Fixes `S-BUG-01`, `S-BUG-03`, `S-BUG-04`)

- **Instant Offline Short-Circuit (`S-BUG-04`):** Inject `ConnectivityService` into `ItineraryService`. If `this.connectivityService.isHostOfflineSnapshot` is `true` (or `!navigator.onLine`), skip the 4-second HTTP timeout and invoke `generateOfflineSuggestedItinerary(request)` in `0ms`.
- **Use `matchedPool` & Respect Daily Budget (`S-BUG-03`):** When `targetDest` specifies a municipality (e.g., *Pantabangan*, *San Jose City*, *Gapan City*, *General Tinio*, *Cabanatuan City*) or when preferences/budget are set, build each day's 3 stops by slicing without replacement from `matchedPool` (filtered by cumulative `request.daily_budget` and Haversine proximity to the day's anchor) instead of falling back to `curatedPool[0..2]`.
- **Honor `placeId` Deep-Link Anchor (`S-BUG-01`):** Add `anchor_place_id?: string` to `SuggestedItineraryRequest`. When `SuggestedWizardPage` receives `queryParams['placeId']`, lock that POI as Stop 1 of Day 1 and cluster the remaining Day 1 stops around its coordinates.

Proposal 3: Synchronize `CustomBuilderPage` Date Changes, Category Mapping & Step 1 Controls (Fixes `C-BUG-01`, `C-BUG-02`, `C-BUG-03`, `C-BUG-05`)

- **Add Title Input & Quick Chips to Step 1 (`C-BUG-01`):** Add the `Trip Title` input box and the reusable Quick Destination & Budget preset chips to `custom-builder.page.html` (`activeView === 'form'`), and change the primary CTA text from *"Find Itinerary"* to *"Start Building Itinerary"*.
- **Reactive Date-to-Days Synchronization (`C-BUG-05`):** Subscribe specifically to `start_date` and `end_date` value changes in `CustomBuilderPage.ngOnInit()` and invoke a non-destructive `syncDaysWithDateRange()` helper that updates every `day.travel_date = toLocalIsoDate(addDaysLocal(start, idx))` and appends/trims days while preserving existing stops.
- **Complete 16-Category Filter Mapping (`C-BUG-02`):** Expand `browseCategories` in `CustomBuilderPage` to cover all 16 canonical categories: `All`, `Attractions & Nature`, `Food & Cafes`, `Resorts & Stays`, `Culture & Heritage`, `Shopping & Markets`, and `Transit & Essentials` (mapping terminals, hospitals, pharmacies, ATMs, fuel stations, convenience stores, and public facilities).

Proposal 4: Unify the Dual Itinerary Editors & Fix Modal Tab Routing (Fixes `UC-C4`, `UC-C5`, `S-BUG-06`)

- **Direct Tab Targeting on Place Picker Modal:** Update `openPlacePicker(defaultTab: 'browse' | 'custom' = 'browse')` in `custom-builder.page.ts:732` and bind the *Add Off-Grid Custom Venue* button to `openPlacePicker('custom')`. Remove `[disabled]="isPlaceInActiveDay(place.id)"` on `.picker-add-action-btn` so clicking an already-added place triggers the helpful duplicate-stop confirmation alert (`selectPlaceFromPicker`) as designed.
- **Draft State Handoff for *Customize in Builder* (`S-BUG-06`):** Instead of calling `createItinerary()` before entering `/custom-builder`, pass `this.generatedPlan` via `this.router.navigate(['/custom-builder'], { state: { draftItinerary:

this.generatedPlan } })` so the trip is only persisted when the user explicitly clicks **Save**.

- **Unified Editor Experience (`UC-C5`):** On `ItineraryDetailPage`, wire the primary **Edit** action to open `/custom-builder/:id` (or embed the Place Picker modal and `pendingChangesGuard` directly into `ItineraryDetailPage` while preserving separate `estimated_cost_min` and `estimated_cost_max` fields).

5.5 Prioritized Implementation & Verification Roadmap

Phase	Target Files	Key Engineering Changes	Verification Gate
Phase 1 Data Integrity & Offline Fixes (P0)	`custom-builder.page.ts`, `itinerary.service.ts`, `suggested-wizard.page.ts`	Fix `C-BUG-06` (preserve `place` & `custom_title` in Guest Custom save); fix `S-BUG-03` (wire `matchedPool` & budget filter in offline generator); fix `S-BUG-04` (0ms offline check); fix `S-BUG-02` (local date helpers).	Unit tests in `itinerary.service.spec.ts` & `custom-builder.page.spec.ts` asserting non-Bongabon offline generation & stop title/coord retention.
Phase 2 Form, Browse & Date Sync (P1)	`custom-builder.page.html`, `custom-builder.page.ts`	Add `Trip Title` & quick chips to Step 1 (`C-BUG-01`); map all 16 categories in `catMap` (`C-BUG-02`); add `syncDaysWithDateRange()` on timeline date changes (`C-BUG-05`); fix `openPlacePicker('custom')`.	`npm run test:unit` & Playwright test verifying Step 1 title input, 16-category filtering, and timeline date resizing.
Phase 3 Unified Editor & Route Map (P2)	`itinerary-detail.page.ts`, `itinerary-detail.page.html`	Unify editor flows (preserving `min/max` cost ranges & adding `pendingChangesGuard`), pass draft state on `customizeInBuilder()`, and redirect post-save to `/itinerary/:id`.	Full CI verification: `npm run lint`, `npm run build` (0 SCSS budget warnings), and E2E journey pass.